Sarvesh Patham

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Career Objective

To improve industry experience in robotics, sensor processing, and analog board design through a summer internship

Education

University of Illinois Urbana Champaign — Urbana, IL
Bachelor of Sciences in Electrical Engineering, Minor in Computer Science — *Aug. 2024 – May 2028*

- Cumulative GPA: 4.0
- Relevant Coursework: Analog Signal Processing, Circuit Analysis and Design, Computer Architecture, Discrete Mathematics, Computer Systems

Experience

Autonomy Engineer — June 2025 – Present
General Autonomy AI — *Piscataway, NJ*

- Developed LiDAR/Camera/IMU data collection stack using NVIDIA Jetson Orin for simulator development
- Implemented several ROS2 nodes using python for data processing including obstacle detection and 3D mapping
- Pitched stack to potential and current investors using a live demonstration of the data gathered

Undergraduate Research Assistant — August 2025 – Present
University of Illinois Urbana Champaign — *Urbana, IL*

- Building mapping algorithms in python for matching simulated magnetic field lines to prototype portable MRI
- Designed hall effect sensor mount and transceiver + pre-amp setups using Fusion360
- Researching transceiver designs for decreasing signal-to-noise ratio and increasing data clarity for low field MRI

Engineering Intern — October 2023 – April 2024
3D Pets — *Bloomingdale, NJ*

- Machined/constructed prosthetics to specific consumer standards; increased production rate by 10 percent
- Manufactured body molds and developed 3D Models for prosthetics by scanning molds with LiDAR scanner
- Developed an organizational standard for parts for a newly constructed workshop to efficiently increase workflow

Research Intern — September 2023 – March 2024
NJIT - STEM for Success — *Remote*

- Researched electroencephalograms (EEGs) with NJIT Professors Dr. Lipuma and Dr. Bukiet
- Composed an introductory article discussing EEGs and their possible functions in prosthetics
- Presented information at the NJIT STEM for Success conference in front of virtual audience

Projects/Extracurriculars

Neurotech | *Signal Processing Engineer* — February 2025 – Present

- Processed EEG data from the frontal lobe/prefrontal cortex by coding digital filters for use in a VR environment
- Filtered data using signal processing techniques including Fast Fourier Transform and wavelet decomposition
- Cleaning noise from signals using filters including notch, bandpass, and high/low frequency filters

Cuff-Link | *Project* — December 2023 – May 2024

- Designed a muscle and motion tracking device using Fusion360 and developed in the Arduino IDE
- Programmed a virtual cursor controlled by onboard IMU and Electromyography sensor (EMG)
- Selected as a top 3 class finalist for school competition; demonstrated product to audience of 400+
- Gathered data using multiple test subjects to tune cursor for improved accuracy

Illinois Space Society | *Hardware Engineer* — October 2024 – February 2025

- Developed a board that manages current and voltage monitoring and high-resolution analog to digital converters (ADC) using KiCAD

Technical Skills

Digital: Java, Python, ROS2, FEM, Linux, C/C++, HTML/CSS, CAD (Fusion360/OnShape), Microcontrollers (Arduino, ESP32)
Hardware: Electrical Design, KiCAD, Electronic Simulation (LTSpice), Soldering